Elite Estate: Local Setup Guide

Follow these steps to get the full Elite Estate stack running on your local machine.

Prerequisites

Ensure you have the following installed:

- **Docker & Docker Compose**: For running the entire stack.
- **Ollama**: For local AI embeddings (runs on your host machine).
- **ngrok account**: Required for Telegram bot webhooks and secure external access.

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1. Local AI Embeddings (Ollama)

Ollama runs on your host machine to leverage your system's performance for generating vector embeddings.

1. **Install Ollama**: Download from [ollama.com (https://ollama.com/)](https://ollama.com/).
2. **Install Model**: Run the following in your terminal to download the embedding model:

   ```
   ollama pull nomic-embed-text
   ```

3. **Cross-Container Access**:
   - On Windows, ensure Ollama is running in the background.
   - If you encounter "Connection Refused" errors from the backend, set the system environment variable `OLLAMA_HOST=0.0.0.0` and restart the Ollama application.

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2. Setting up n8n & External Access (ngrok)

To allow Telegram to communicate with your local n8n instance and access the n8n editor via the web, you must set up a public tunnel.

Step A: Create an ngrok Free Domain

1. Sign up at [ngrok.com (https://ngrok.com/)](https://ngrok.com/).
2. Navigate to **Cloud Edge > Domains**.
3. Click **Create Domain** (one static domain is free per account).
4. Copy your domain (e.g., `your-name.ngrok-free.app`).

Step B: Configure n8n in .env

1. Retrieve your `NGROK_AUTHTOKEN` from the ngrok dashboard.
2. Update your `.env` file with these values:

   ```
   NGROK_AUTHTOKEN=your_token_here
   N8N_HOST=your-name.ngrok-free.app
   WEBHOOK_URL=https://your-name.ngrok-free.app
   N8N_EDITOR_BASE_URL=https://your-name.ngrok-free.app
   ```

3. **Note**: Once the stack is running, you will access the n8n dashboard using your ngrok link: `https://your-name.ngrok-free.app`.

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3. External Services Configuration

3.1 ImageKit.io (Media Management)

ImageKit handles property images via a global CDN.

1. Register at [imagekit.io (https://imagekit.io/)](https://imagekit.io/).
2. In the dashboard, go to **Developer Options**.
3. Copy the following into your `.env`:
   - `IMAGEKIT_PUBLIC_KEY`
   - `IMAGEKIT_PRIVATE_KEY`
   - `IMAGEKIT_URL_ENDPOINT` (e.g., `https://ik.imagekit.io/your_id/`)

3.2 Google Cloud Console (Calendar API)

Required for synchronizing property visits with agent calendars.

1. Go to [Google Cloud Console (https://console.cloud.google.com/)](https://console.cloud.google.com/).
2. **Create a Project**: Select "New Project" and give it a name.
3. **Enable API**: Search for "Google Calendar API" and click **Enable**.
4. **OAuth Consent Screen**:
    - Go to **APIs & Services > OAuth consent screen**.
    - Choose "External" and fill in the required app information and support email.
5. **Create Credentials**:
    - Go to **APIs & Services > Credentials**.
    - Click **Create Credentials > OAuth client ID**.
    - Application type: **Web application**.
    - **Authorized Redirect URIs**: Add `https://your-name.ngrok-free.app/rest/oauth2-callback`.
6. Copy the **Client ID** and **Client Secret** (you will use these in the next step).

3.3 Connecting Google Calendar in n8n

Once n8n is running, you must link it to your Google account.

1. Open your n8n editor (`https://your-name.ngrok-free.app`).
2. Open the **Smart Agent** or **Reminder** workflow.
3. Click on any **Google Calendar node**.
4. In the **Credential** dropdown, select **Create New Credential**.
5. Authentication Method: Select **OAuth2**.
6. Enter the **Client ID** and **Client Secret** from the Google Cloud Console.
7. Click **Sign in with Google** and authorize the application.

3.4 OpenRouter (AI Agent)

1. Sign up at [openrouter.ai (https://openrouter.ai/)](https://openrouter.ai/).
2. Create an API Key and add it to `.env` as `OPENROUTER_API_KEY`.
3. In the n8n **Smart Agent** workflow, ensure the model is set to `google/gemini-2.0-flash-001`.

3.5 Telegram Bot

1. Message [@BotFather (https://t.me/botfather)](https://t.me/botfather) to create a new bot.
2. Copy the **API Token** into your `.env`.

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4. Launching the Application

1. **Prepare Environment**:

```
cp .env.example .env
# Ensure all keys from the steps above are filled in.
```

2. **Start the Stack**:

```
docker-compose up -d --build
```

3. **Initialize the Database**:

```
docker exec -it FastAPI_backend python seed.py
```

Access Points:

- **Frontend Dashboard**: [http://localhost:3000 (http://localhost:3000)](http://localhost:3000)
- **Interactive API Documentation**: [http://localhost:8000/docs (http://localhost:8000/docs)](http://localhost:8000/docs)
- **n8n Automation Editor**: `https://your-name.ngrok-free.app` (Using your ngrok tunnel)

Troubleshooting

- **Ollama Error**: Run `ollama list` to verify `nomic-embed-text` is installed.

- **Telegram Webhook**: If the bot doesn't respond, check the `WEBHOOK_URL` in `.env` and ensure it uses `https://`.
- **Google OAuth**: Ensure the redirect URI in Google Console exactly matches the one in your `.env` (including the `/rest/oauth2-callback` suffix).